The broken stick problem as the continuum limit of a Bernoulli process

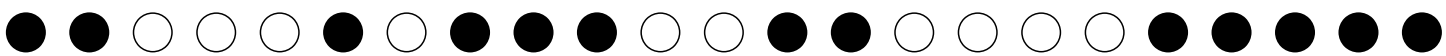
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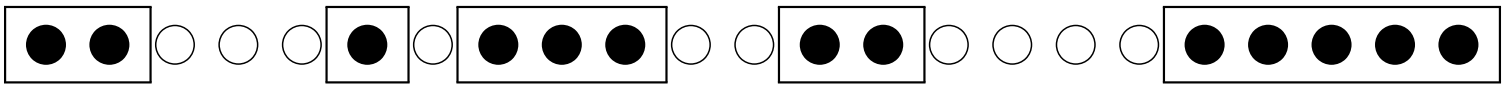
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Run

A **run** is consecutive identical elements in a sequence.
Sequence of Bernoulli trials:



Run of "●":



The length of the longest run is 5.

The Research Question

Behavior of the longest run when sequence length $n \to \infty$ while the probability of getting a perfect sequence is constant:

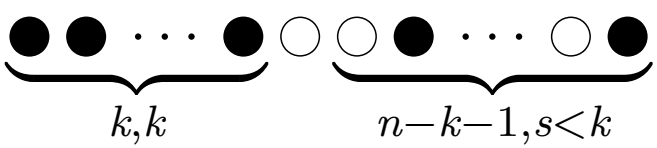
$$n \to \infty,$$

$$p^n = P = \text{const.}$$

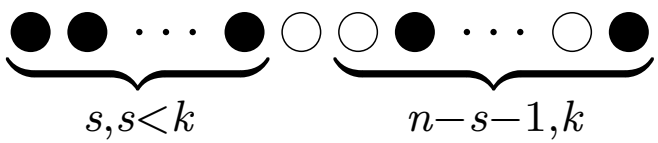
The Discrete Model

Consider a Bernoulli sequence of length n, where each element is either success (S) or failure (F), with probabilities p and $1-p$ respectively, and independent of each other. Let $P_{n,k}(p)$ be the probability that the length of the longest run of S's is k. What is the expression of $P_{n,k}(p)$?

The Recurrence Relation



$$p^k (1-p) \sum_{s=0}^{k} P_{n-k-1,s}$$

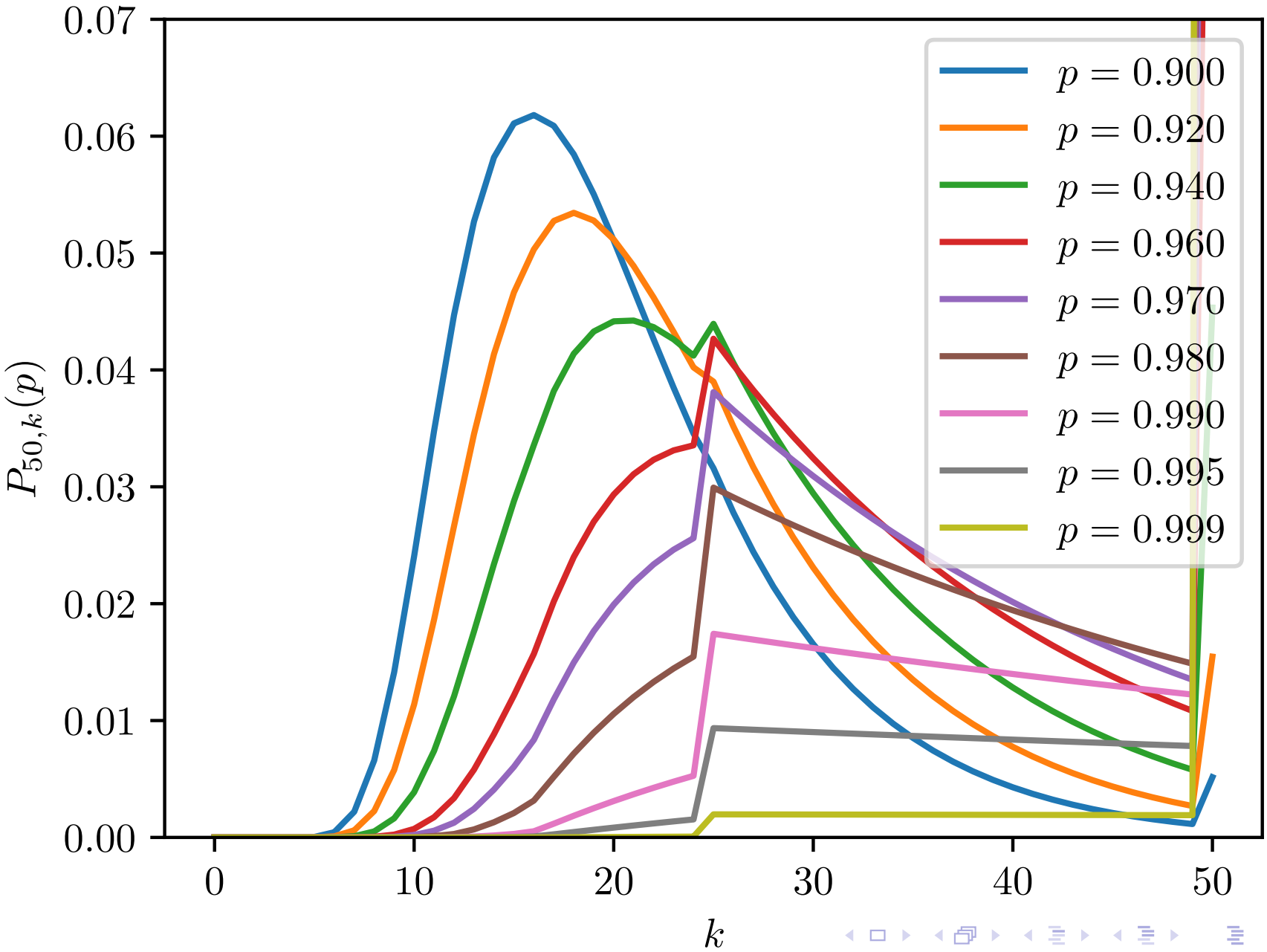


$$\sum_{s=0}^{k-1} p^s (1-p) P_{n-s-1,k}$$

$$P_{n,0<k<n} = (1-p)\left(p^k \sum_{s=0}^{k} P_{n-k-1,s} + \sum_{s=0}^{k-1} p^s P_{n-s-1,k}\right).$$

$$P_{n,0} = (1-p)^n, \qquad P_{n,n} = p^n, \qquad P_{n,k>n} = 0.$$

Visualizing the Discrete Distribution



Defining the Continuum Limit

Define $P := p^n$ and keep it constant as $n \to \infty$. The probability distribution of $x := k/n$ is

$$f(P, x) := \lim_{n\to\infty} nP_{n,xn}\left(P^{1/n}\right) \qquad (0 \leq P \leq 1, \quad x \geq 0)\,.$$

The Integral Equation

$$g(P,x) := \frac{1}{P} f(P,x) - \delta(x-1)\,.$$

$$g(P, 0 \le x \le 1) = -\ln P \left(\int_0^{x/(1-x)} \left(g\left(P^{1-x}, u\right) + \delta(u-1) \right) \mathrm{d}u \right.$$
$$\left. + \int_x^{x/(1-x)} \left(g\left(P^{x/v}, v\right) + \delta(v-1) \right) \frac{\mathrm{d}v}{v} \right),$$

$$g(P, x > 1) = 0.$$

$$\int_0^\infty g(P,x)\,\mathrm{d}x = \frac{1}{P} - 1.$$

Overview of the ADM

The Adomian decomposition method (ADM):

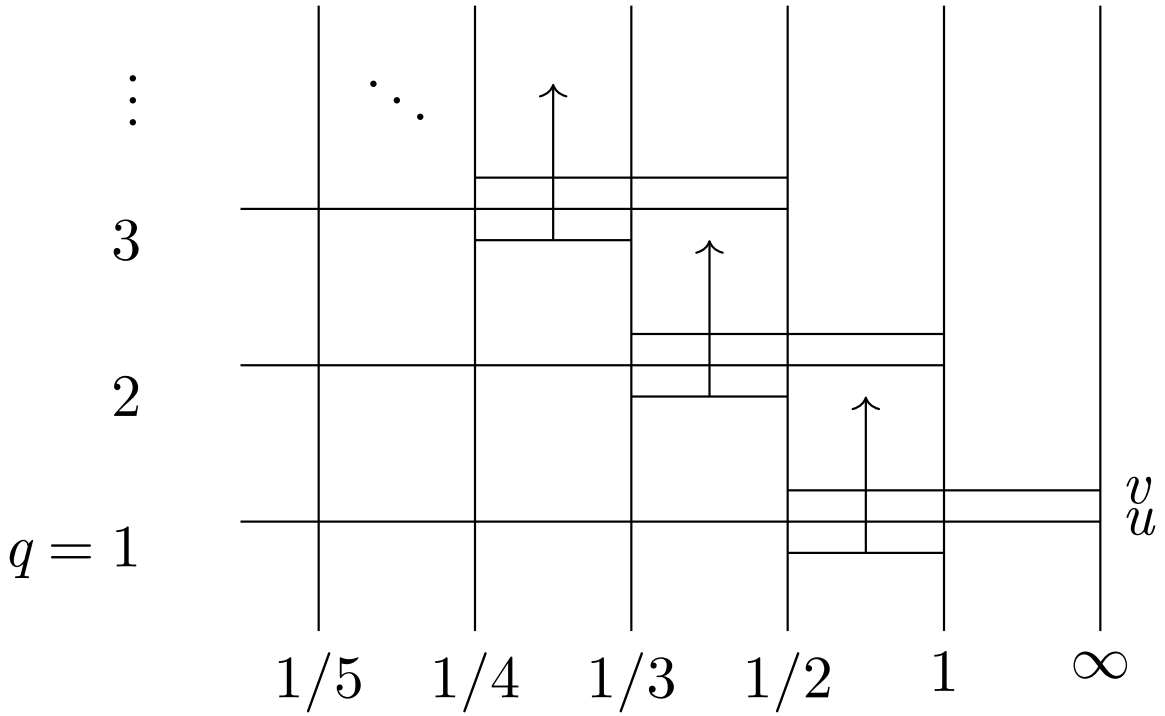
$$g = \Phi(g)\,.$$

$$g = g_0 + \lambda g_1 + \lambda^2 g_2 + \lambda^3 g_3 + \cdots\,,$$
$$\Phi(g) = \Phi_0 + \lambda \Phi_1(g) + \lambda^2 \Phi_2(g,g) + \lambda^3 \Phi_3(g,g,g) + \cdots\,.$$

$$g_j = \sum_{l=0}^{j} \sum_{m_1+\cdots+m_l=j-l} \Phi_l(g_{m_1},\ldots,g_{m_l})\,.$$

Piece-by-Piece Solution

$$g^{(q)}(P, x) := g(P, x) \qquad \left(\frac{1}{q+1} < x < \frac{1}{q}\right).$$

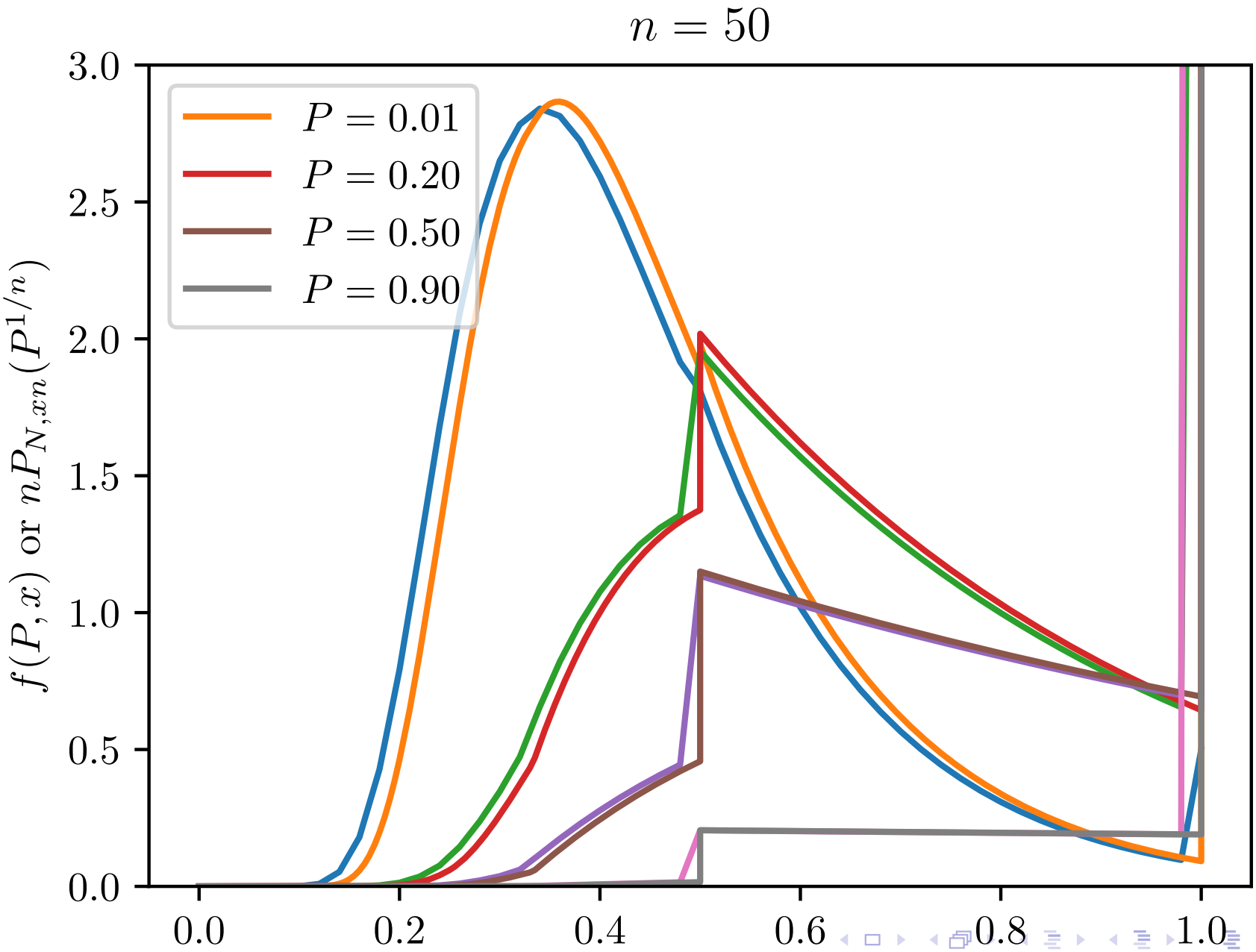


Solution

$$F(P,x) \coloneqq \int_0^x \mathrm{d}x' \int_0^{x'} \mathrm{d}x''\, f\big(P,x''\big)\,, \qquad r_s \coloneqq \ln P^{sx-1}.$$

$$F(P,x) = \frac{P}{\ln P} \sum_{s=1}^{\lfloor 1/x \rfloor} \frac{(-1)^s}{s!\, s} \mathrm{e}^{r_s} r_s^s - \frac{\operatorname{li} P - \gamma - \ln(-\ln P)}{\ln P} + x.$$

Visualization of the Solution



The Broken Stick Problem

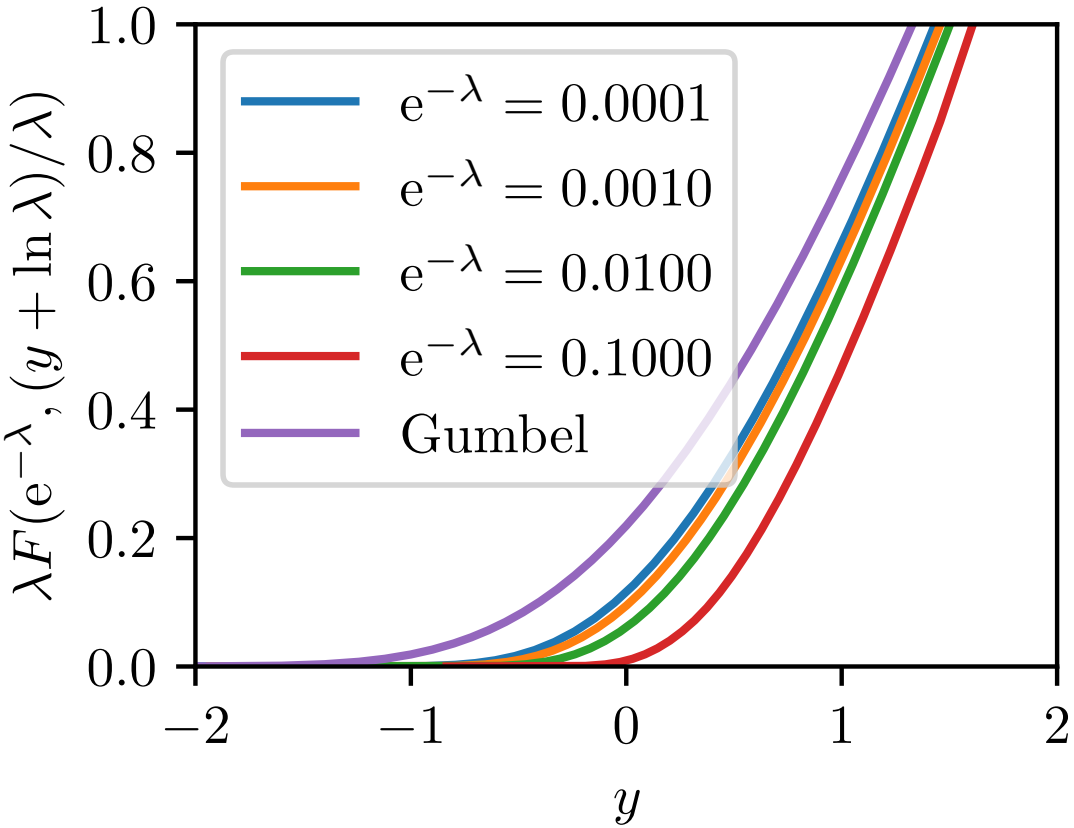
Break a unit-length stick at k random points. The distribution of the longest segment is $f_k(x)$.

$$f\left(\mathrm{e}^{-\lambda}, x\right) = \mathrm{e}^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} f_k(x)\,,$$

$$f(P, x) = P \sum_{k=0}^{\infty} \frac{(-\ln P)^k}{k!} f_k(x)\,.$$

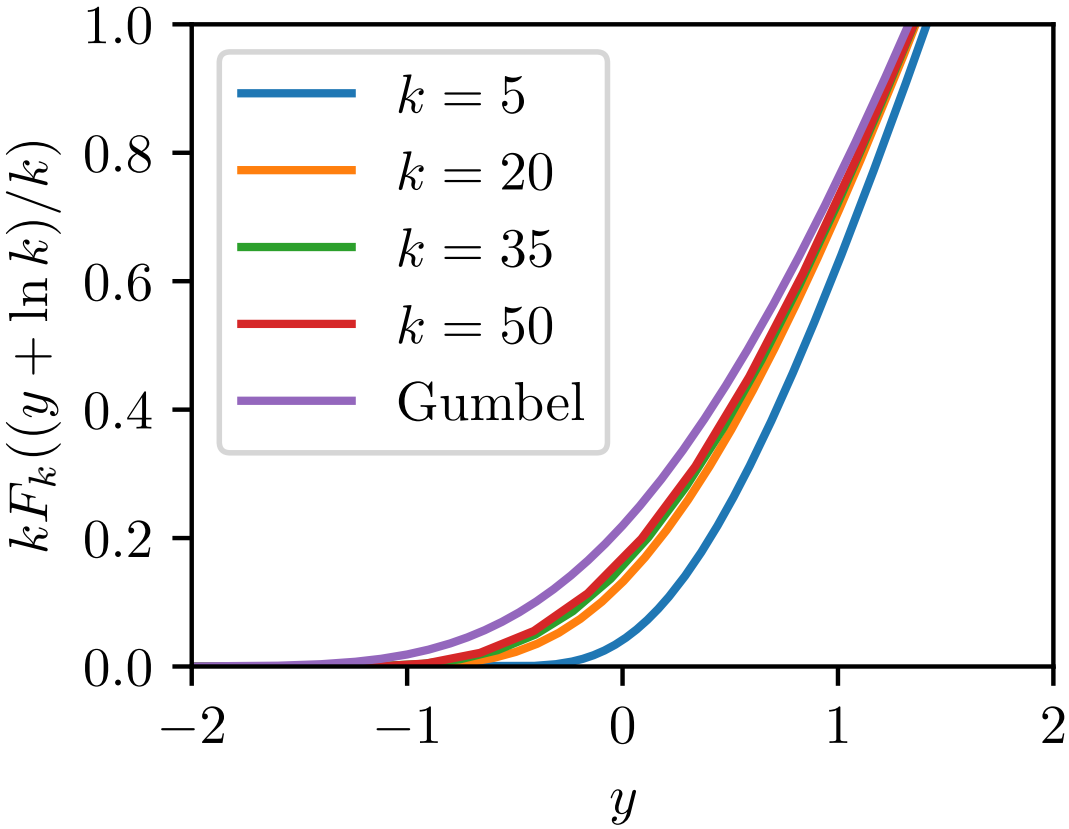
Run Statistics to the Gumbel Distribution

$f(P, x)$ tends to the Gumbel distribution as $P \to 0$.



Broken Stick Problem to the Gumbel Distribution

$f_k(x)$ tends to the Gumbel distribution as $k \to \infty$.



Summary of Findings

- Derive a recurrence relation for the longest run distribution and formulate its continuum limit to an integral equation.
- Derive a piecewise analytical solution using the ADM.
- Formally establish the link between run statistics and the broken stick problem.
- Formally derive the Gumbel distribution in limiting cases.